



جامعة الكفيل
University of Alkafeel

Oral & Maxillofacial Radiology

Introduction

ا.م. سرمد محمد حموزي

A. prof. Sarmad Hamozi

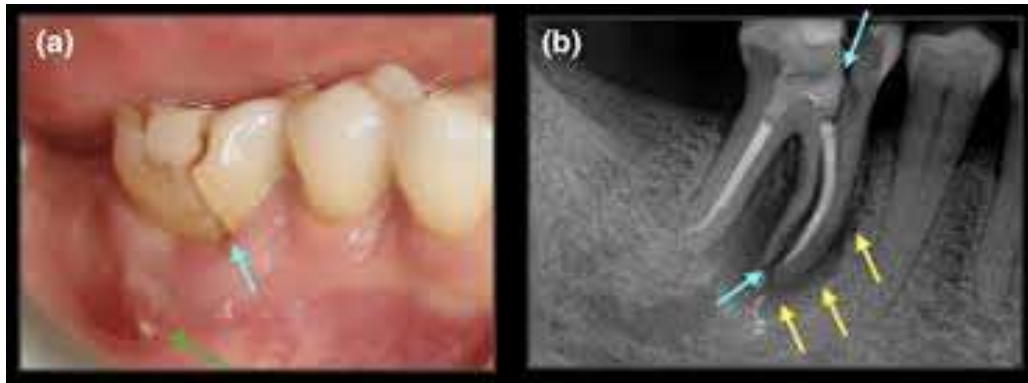
B.D.S, M.Sc.,OMFR, Ortho ADA

dr.sarmadh@alkafeel.com

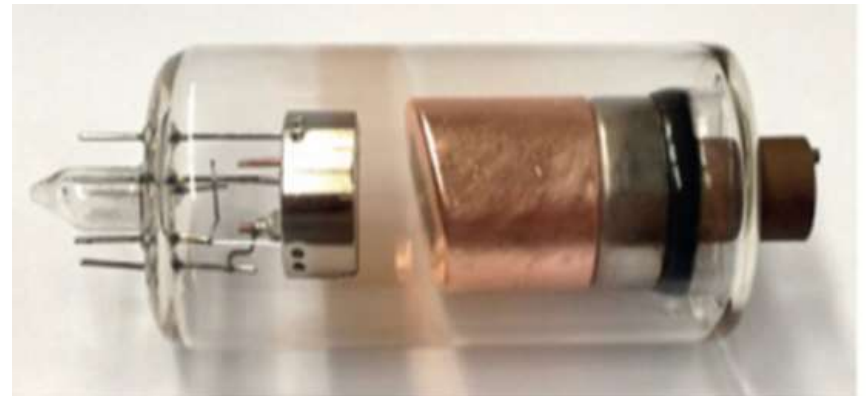
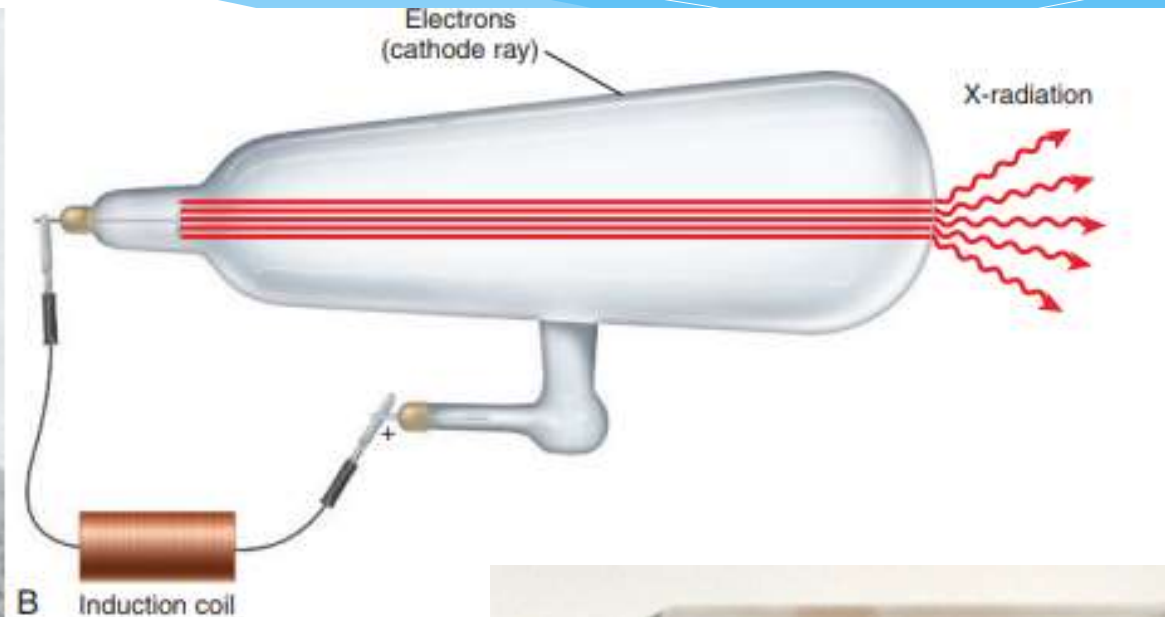


Introduction

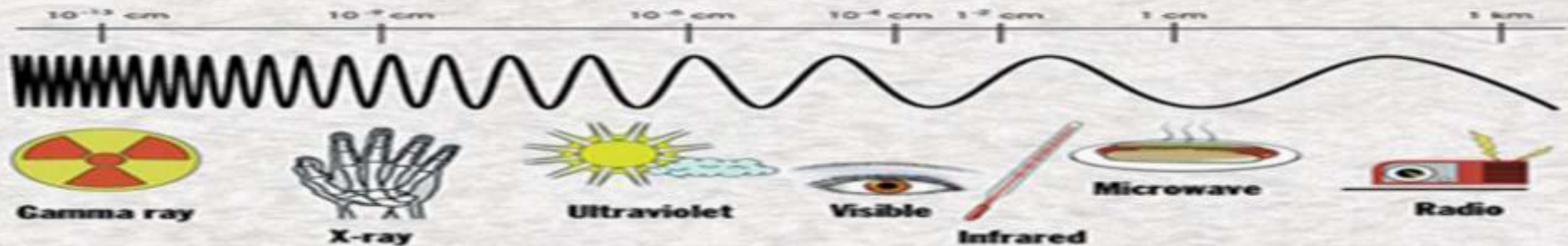
- * Dental radiography is a process of production a photographic image for an object through the use of X – radiation and by this process dentist can examine the internal tissues like (roots of teeth and alveolar bone) which are not able to be seen by the naked eye due to over soft tissues.



X – Ray was discovered by (Roentgen) in 1895, it travels in a form of pure energy and the basic unit is X – ray photon or (quantum).



The Electromagnetic Spectrum



- * X – Ray photons travel with a wave motion called (sine – wave) and the distance between the crests of these waves called (wave – length) which measured by a unit ($\text{\AA}$) Angstrom units.

0,000,000,000,100
m mm μm nm $\text{\AA}$ pm

- * The X – ray photons wave length used in diagnostic radiography is ranged between $0.1 - 0.5 \text{ \AA}$ and the amount of energy contained in each photon called (photon energy) which depend on
- * 1. Wave length
- * 2. Frequency of X – ray

- 
- * The high frequency of X – ray the shorter wave length photons this shorter wave length photon has more energy than a low frequency long wave length type of X – ray photons.

Similarities between X – ray and light

1. Both belong to the same electro – magnetic radiation family.
2. Both travel in straight lines at the same speed which is 186,000 miles per seconds.
3. Both affected the photograph films and made them black.
4. Both not affected by magnetic field



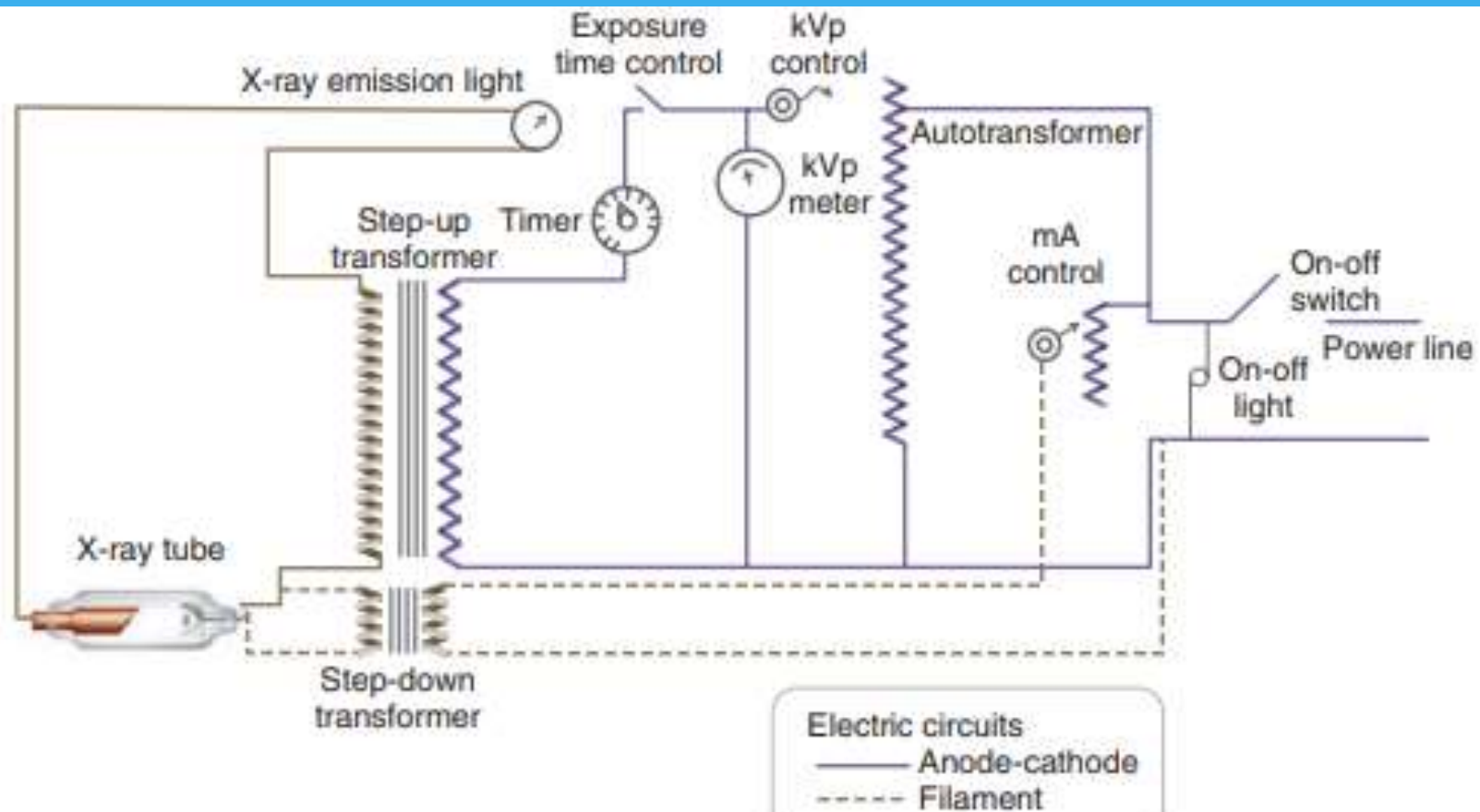
The major difference between visible light and x-ray

1. X– ray has the ability to penetrate objects that the light can not pass through
2. X– ray has the ability to ionize atoms
3. X – ray has the ability to produce light (blue light) when it hits some objects and this phenomena called (fluorescence).
4. X – ray is invisible



The components of X – ray machine





The components of X – ray machine

1. **Step up transformer:** - It consists of 2 coils of electrical wire and it used to increase the incoming voltage from 240 volts to 60 kilovolt or more.
2. **Step down transformer:** - It used to decrease the incoming voltage to around 10 volts in order to supply the filament circuit of X – ray tube with low voltage.
3. **Autotransformer:** - This is made from one coil to do the work of 2 coils. It can be used for making minor change in the voltage.

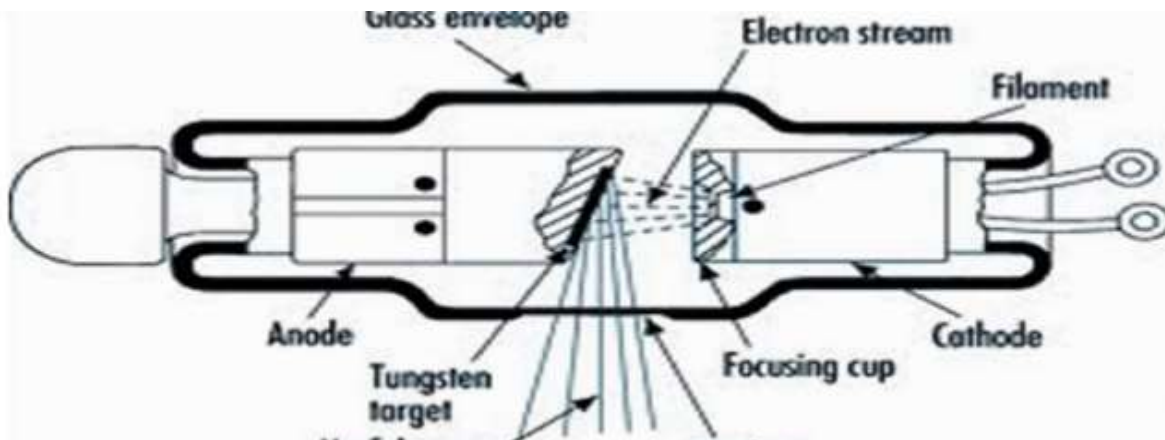
The components of X – ray machine

4. Timer: - It cause activation of high tension current across the tube and this happened when the timer bottom is pressed.

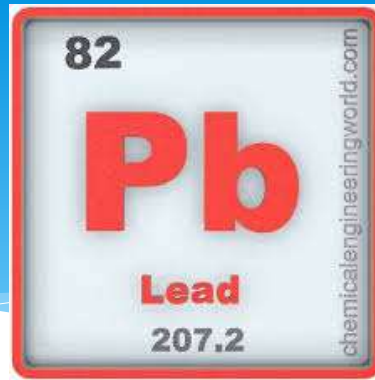
5. Ammeter and Voltmeter: - These two are, is the same type of instrument since both are operated by magnetic field.

The components of X – ray machine

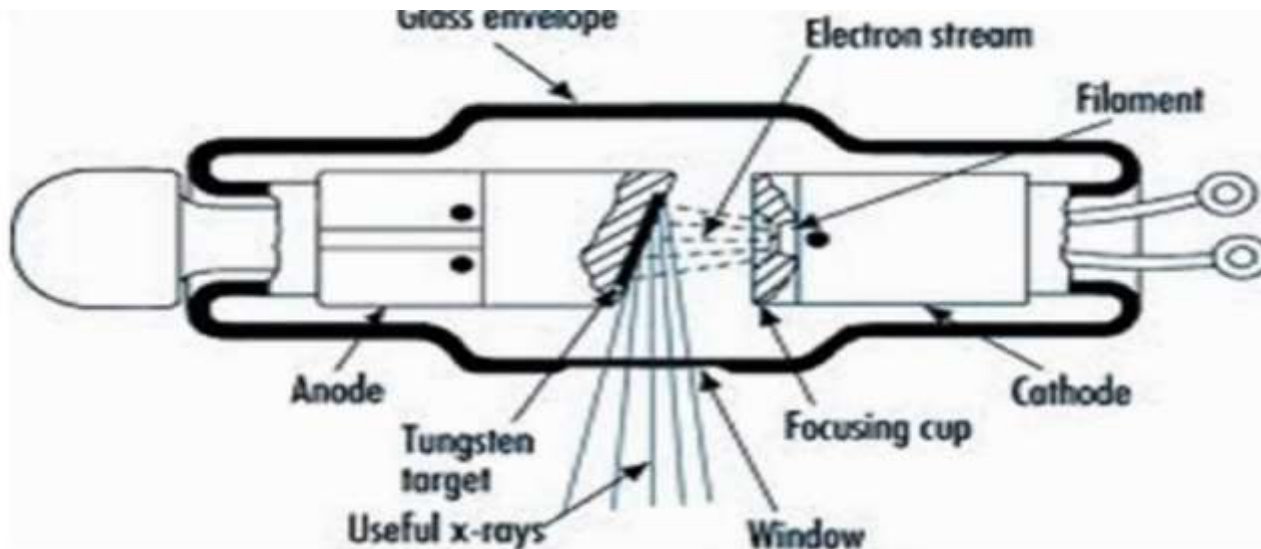
6. **X – ray tube:** - It's a vacuum tube so that the electrons are free to travel inside the tube without interaction with air molecules.







- * The glass the tube is leaded to prevent (the generated X – ray) from escaping in all directions. While the window is of unleaded glass so that X – ray exist out through this window.



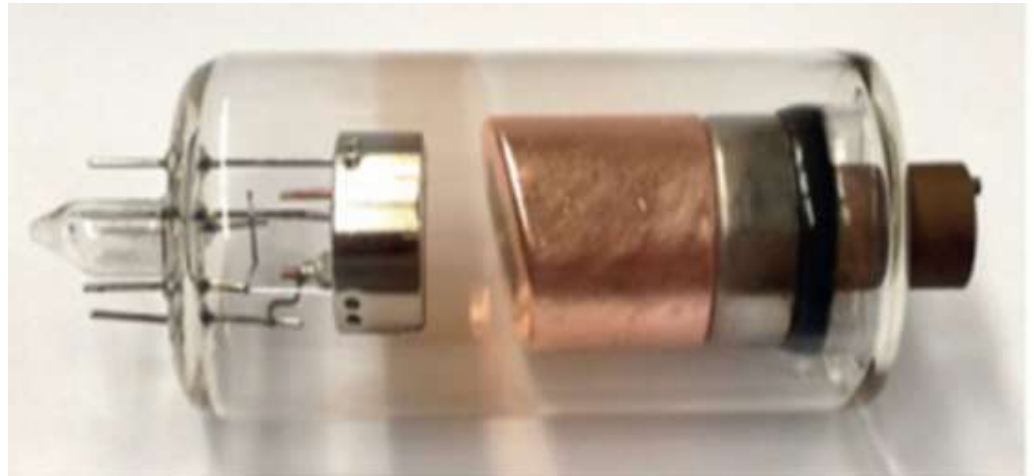
x – ray tube consist of 2 ends

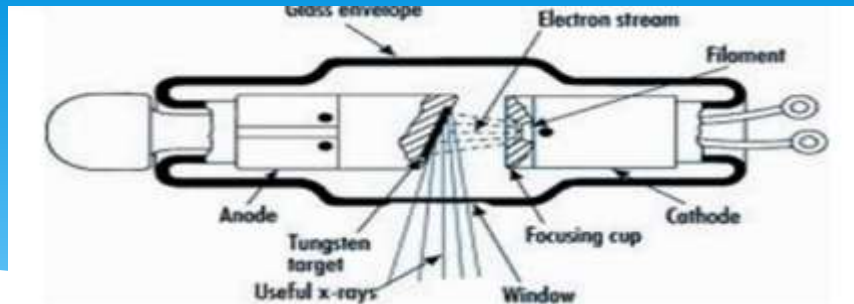
A – (-Ve) electrode (cathode) end.

B – (+Ve) electrode (anode) end.

Cathode end:-

It consists of **focusing cup** made of molybdenum in which a **tungsten filament** is set this filament is the source of electrons that are used to generate X – ray.





Anode end:- *

Here we have a thin tungsten button set in a rod of copper this * rod surrounded by an oil bath which used to absorb heat that created during generation of X – ray.



Generation of X – ray

- * X – ray photons generate at the surface and within the tungsten button (target) of the anode by bombardment of anode with electrons coming from cathode when the filament is heated electrically.
- * The speed of these electrons depends on the potential difference (kV) between cathode and anode of the tube.



- * Electrons impact on (a rectangular area) at the surface of tungsten button of the anode called (focal spot)
- * Now the kinetic energy of these electrons will be converted to X – radiation and heat in fact there is only 0.2% of this energy will be converted to X – ray while the remaining energy goes in form of heat this heat will be dissipated with the aid of radiator that is attached to the anode.

Selection of target material

Ideal target material must possess the following qualities:-

- a- It should have high thermal conductivity
- b- It should have high atomic number
- c- It should have high melting point
- d- It should have low vapor pressure

Tungsten possesses all these requirements, but it has low thermal conductivity which is compensated by the copper rods to conduct the heat outside the X-ray tube.

Focal spot

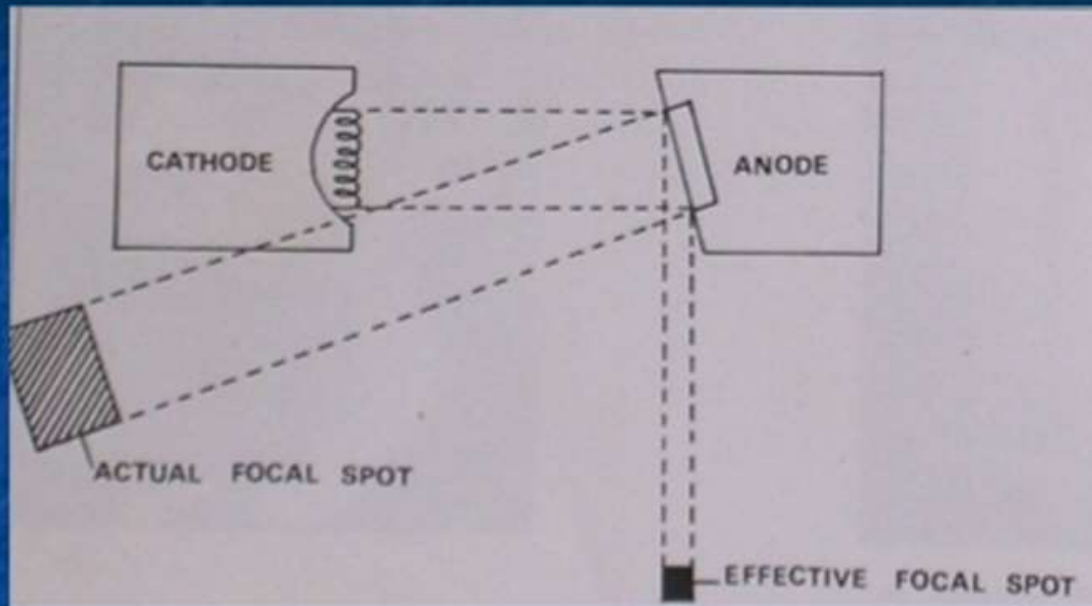
- * It's a small area on the tungsten surface (target) on which electrons come from cathode are impacted and X – ray photons are produced. Therefore useful X – ray beam emerges from this spot the rectangular focal spot called (actual focal spot) while the projection of this spot when viewed from any point with the X – ray beam appears more or less like a square this square projection called (effective focal spot).



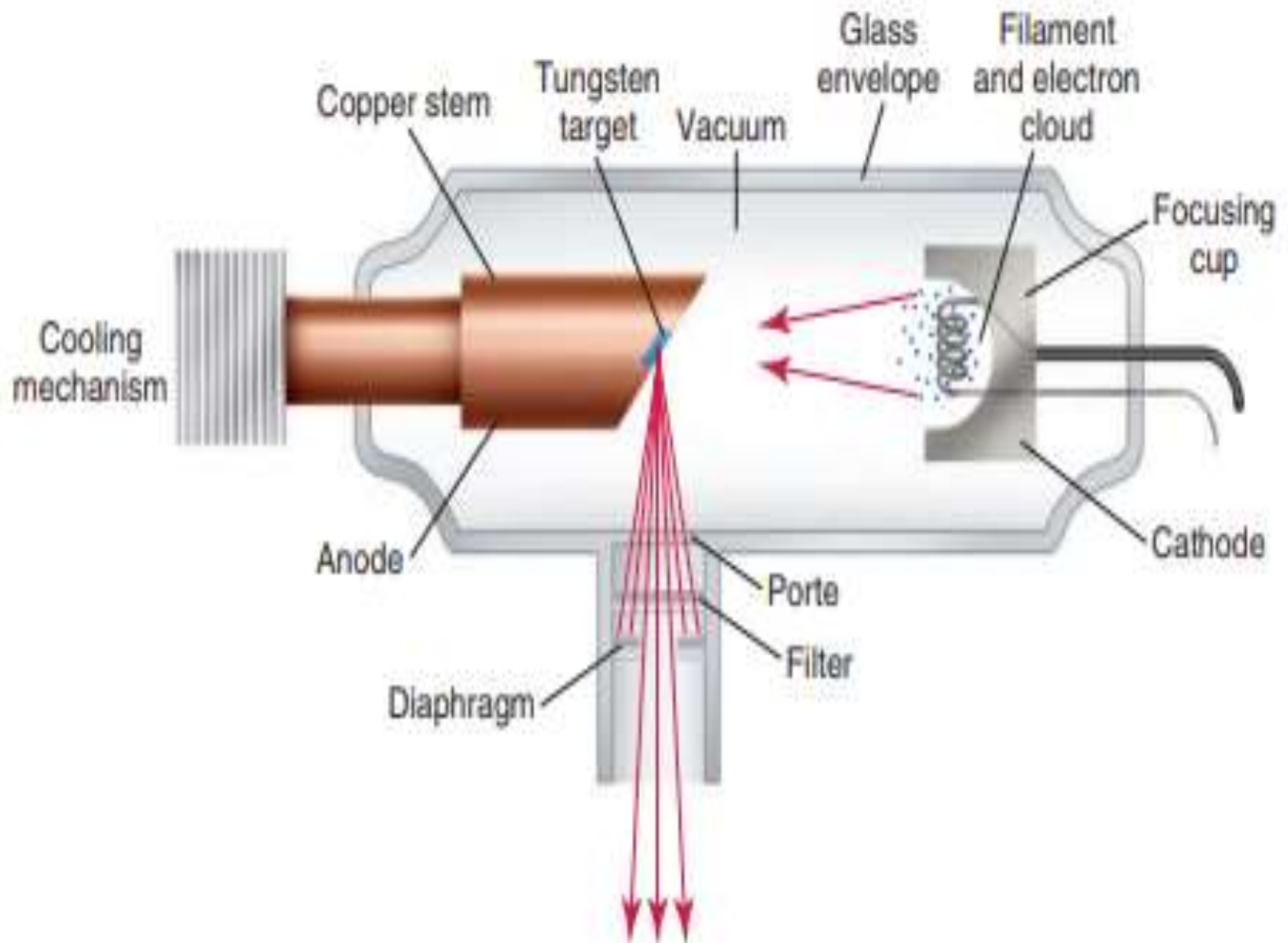
In diagnostic radiography the use of rectangular focal spot which is projected as a square called (Line focus principle) this will be obtained by setting the surface of the target at an appropriate angle $(15 - 20)^\circ$ to the cathode.

This principle helps to get these benefits:-

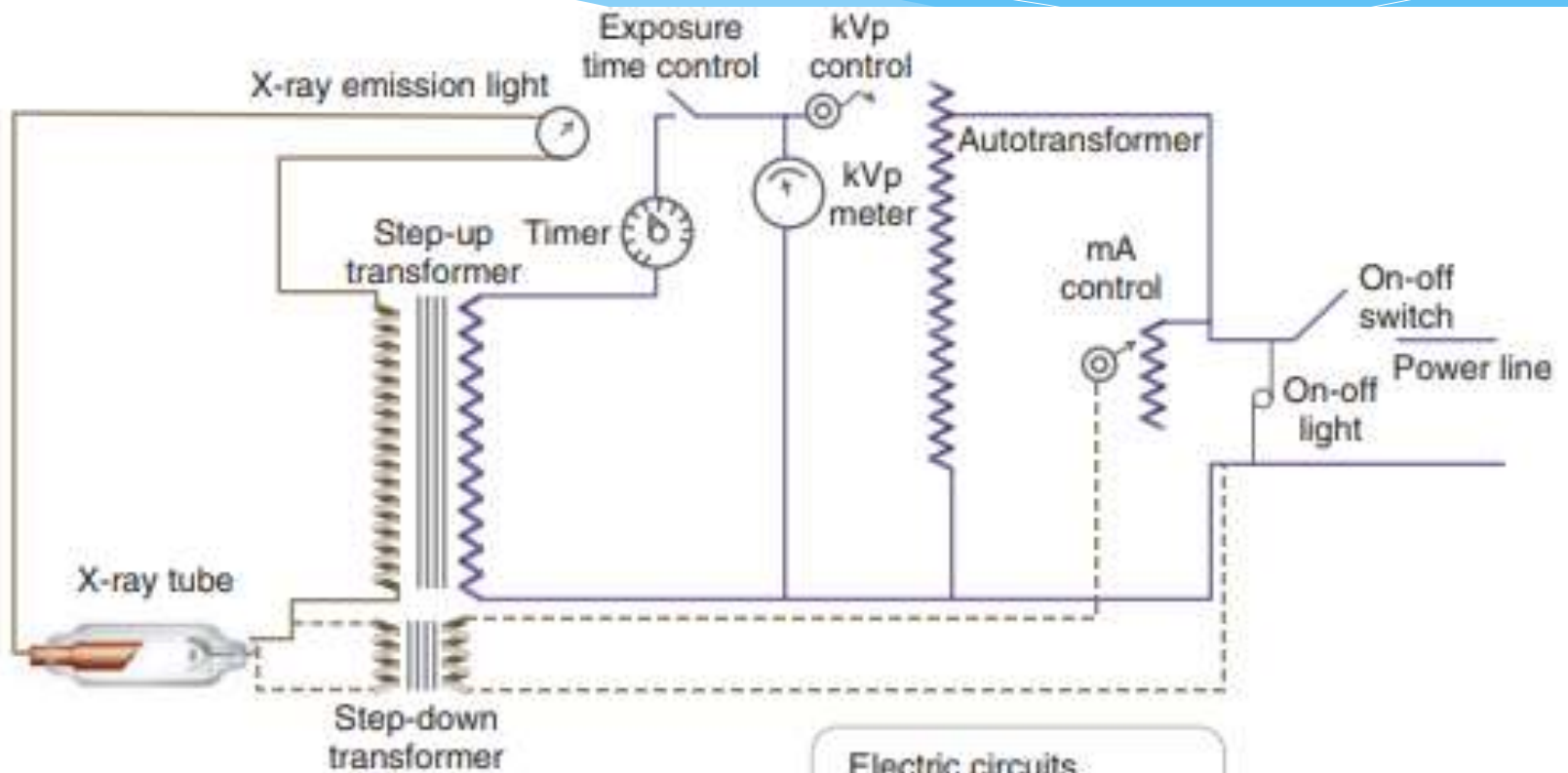
1. X – Ray generated over large surface area on the target.
2. Larger actual focal spot for better dissipation of heat.

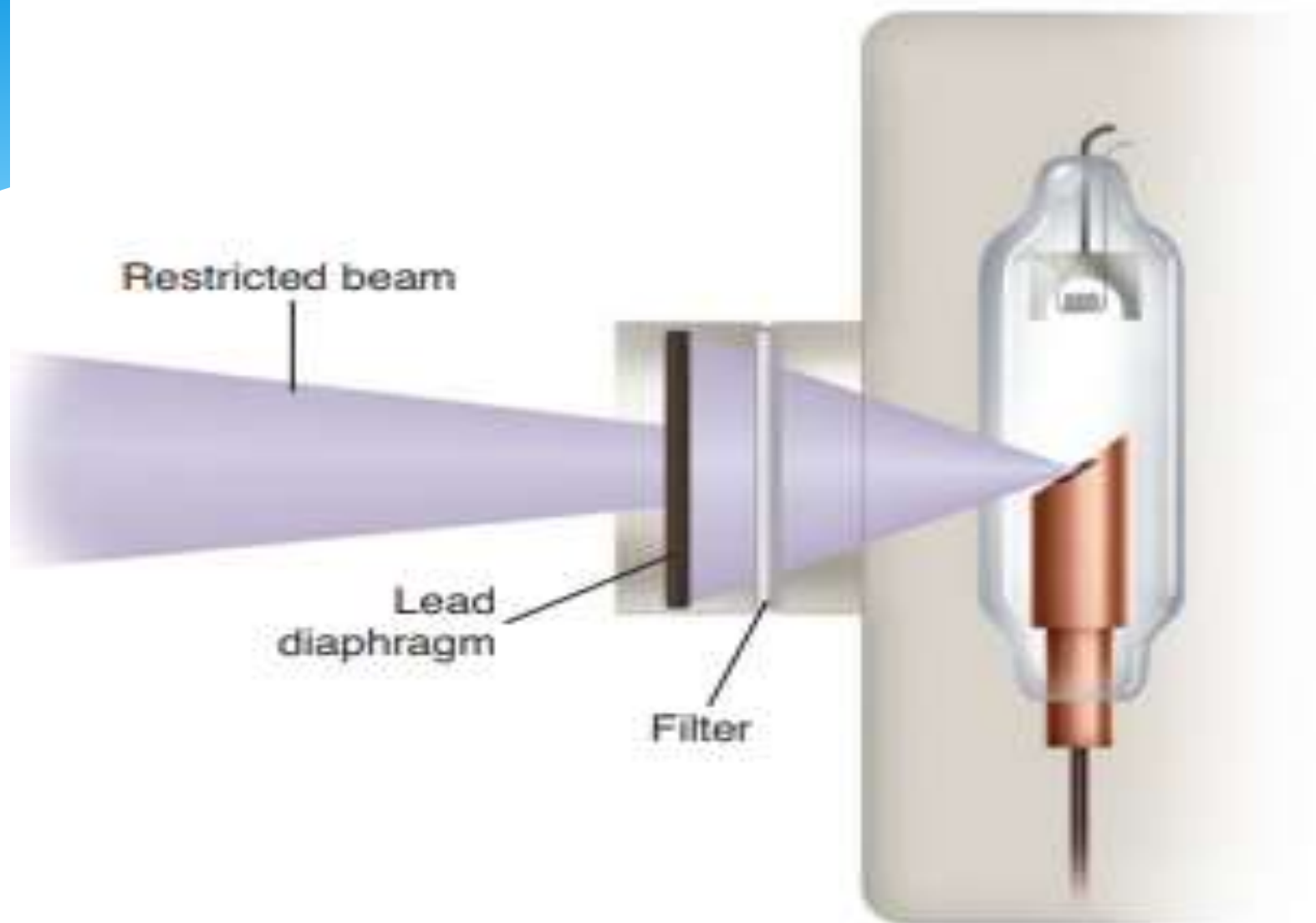


Benson line focus principal.

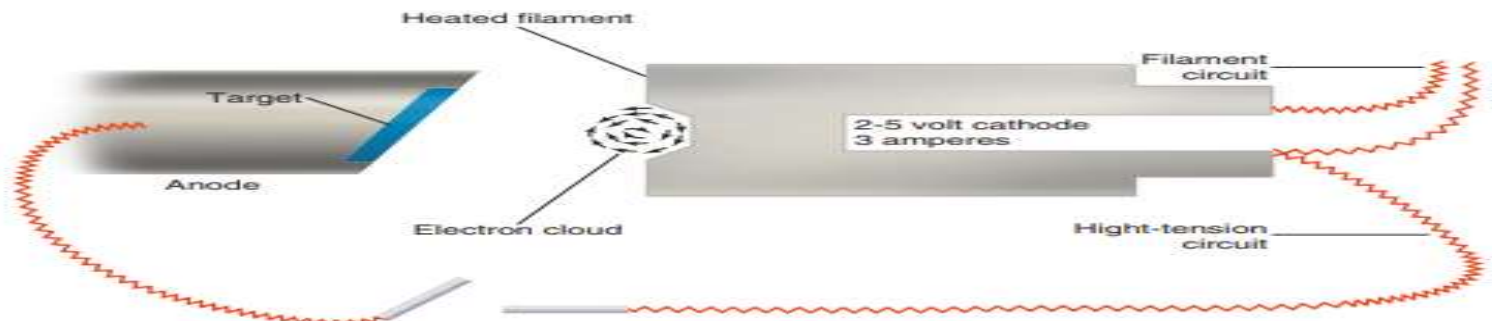


• **Figure 1.12** Components of a dental x-ray tube.

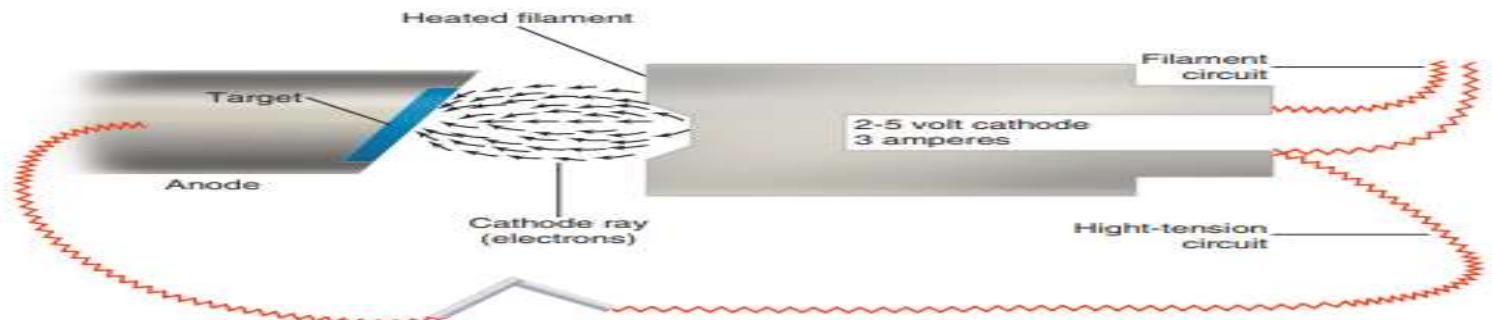




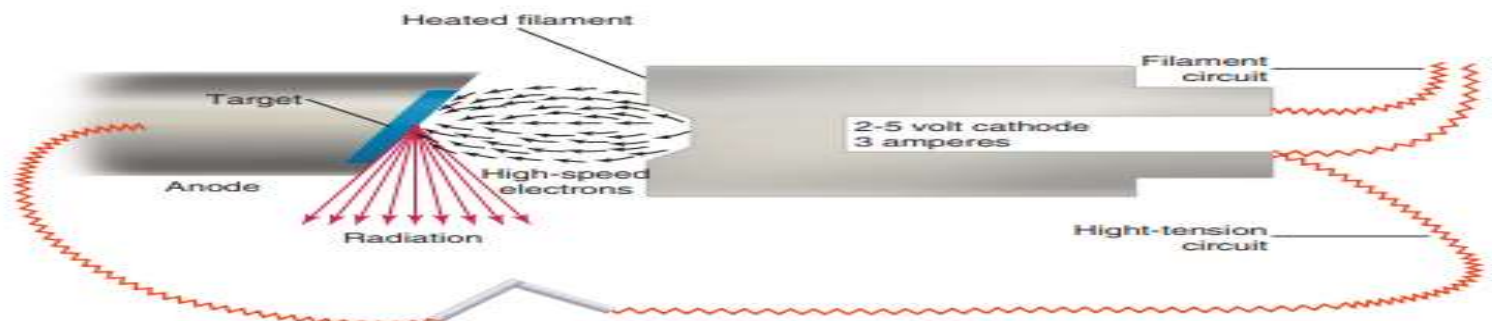
• **Figure 2.12** Collimation and filtration of x-ray beam.



• **Figure 1.14** An x-ray tube illustrating formation of an electron cloud at the cathode as the filament circuit is activated. Note that the exposure switch is open.



• **Figure 1.15** An x-ray tube showing electrons traveling across the tube from the cathode to anode (target) as the high-tension circuit (exposure switch) is activated.



• **Figure 1.16** An x-ray tube showing production of x-rays as high-speed electrons collide with target.

The x-ray beam

Electron - Target interaction

- * Electrons traveling from the filament (cathode) to the target (anode) convert a small percentage (0.2%) of their kinetic energy into x-ray photons by the formation of **bremsstrahlung** and **characteristic** radiation.

A. Bremsstrahlung radiation

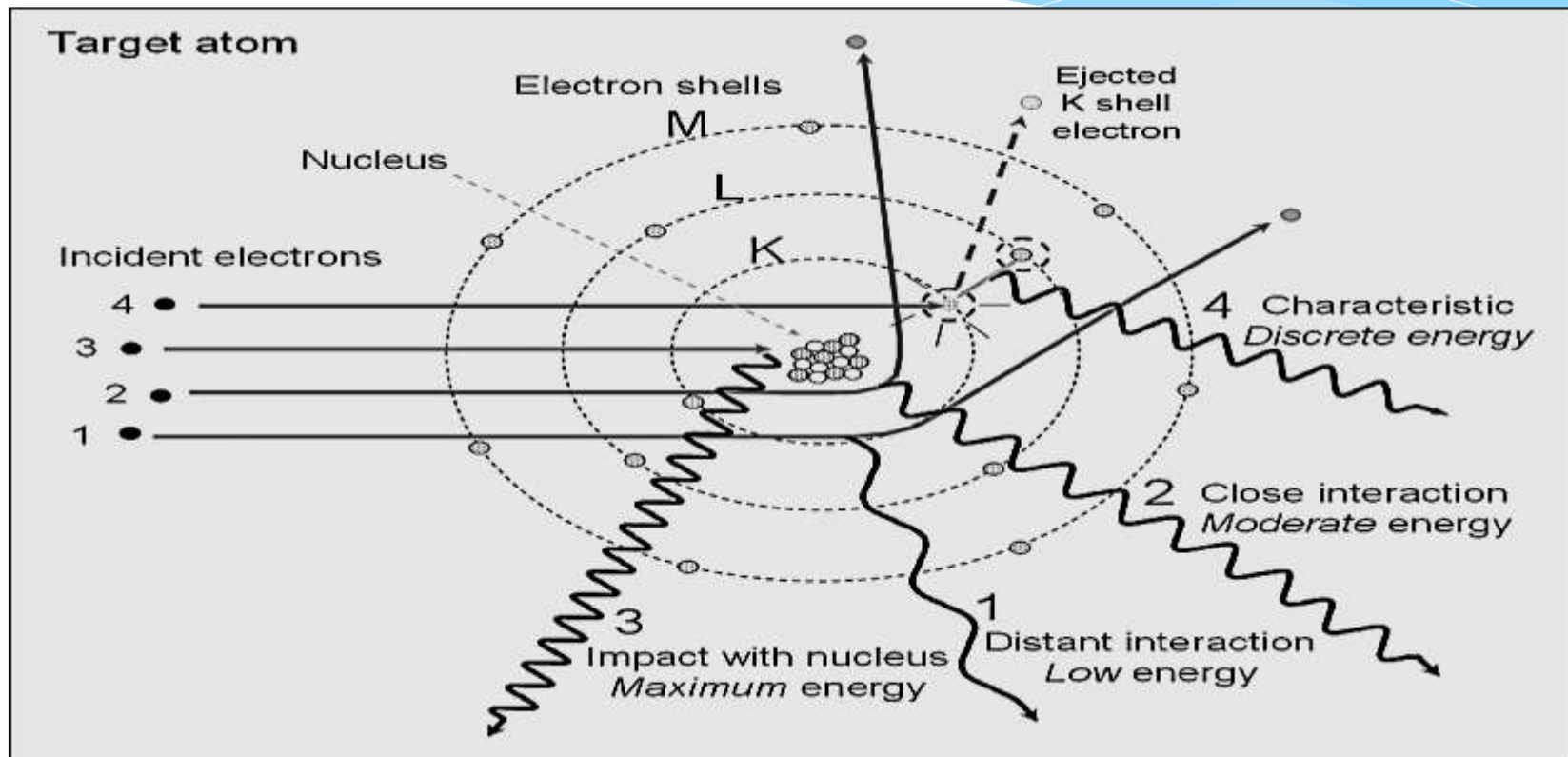
- * the primary source of x-ray photons from an x-ray tube, are produced by the sudden stopping of high-speed electrons at the target. When the electrons from the filament strike the tungsten target,

x-ray photons are created if they either hit a target nucleus directly (rare) or their path takes them close to the nucleus. If a high speed electron hits the nucleus of a target atom, all its kinetic energy is transformed into a single x-ray photon

B. Characteristic radiation

- * Characteristic radiation occurs when an electron from the filament displaces an electron from an inner-shell of the tungsten target atom. When this happens, another electron in an outer-shell of the tungsten atom is quickly attracted into the void in the deficient inner-shell. When the displaced electron is replaced by the outer-shell electron, a photon is emitted with an energy equivalent to the difference in the two orbital binding energies.

So, X- ray beam consist of many photons of different wave length **because: Electrons don't give up all their kinetic energy in identical fashion.**



Factors controlling the x-ray beam

I. Exposure Time

Changing the time controls the duration of the exposure and thus the number of photons generated. When the exposure time is doubled, the number of photons generated is doubled, but the range of photons energies is unchanged.

* II. Tube Current (mA)

The quantity of radiation produced by an x-ray tube (i.e., the number of photons that reach the patient and film) is directly related to the tube current (mA) and the time the tube is operated. The number of x-rays produced depends directly upon the number of electrons that strikes the target. The number of electrons is directly related to the tube current. In dental radiography use of 7-15 mA is required.

A setting above 15 mA is not recommended because of excessive heat production in the x-ray tube.

III. Tube Voltage (kVp)

Increasing the kVp increases the potential difference between the cathode and anode, thus increasing the energy of each electron when it strikes the target. The greater the potential difference the faster the electrons travel from the cathode to the anode. This results in an increased efficiency of conversion of electron energy into x-ray photons, and thus increase the number and energy of photons.

Voltage Rectification

. The main supply to the X – ray machine of 240 volts has

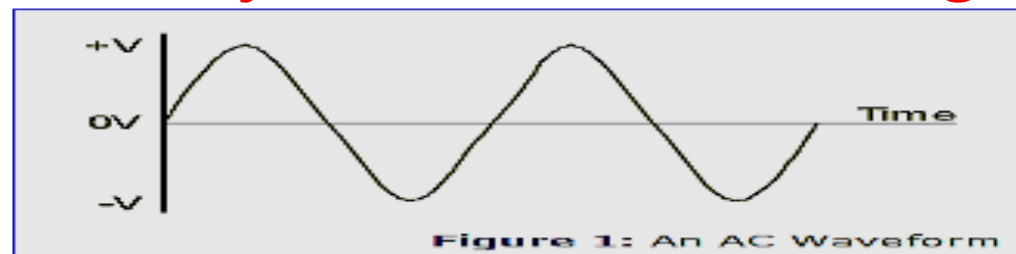
2 functions:

a – Generate the high potential difference (kV) to accelerate the electrons across the X – ray tube via the **step up transformer**.

b – Provide the low – voltage current to heat the filament via the **step – down transformer**.

However, the incoming 240 volts is an alternating current with typical wave form as shown:-

Half of cycle is (+) and other half is (-) but for production of X – ray only the positive half of the cycle can be used because x-ray produced by acceleration of electrons from cathode to anode and cannot be produced by electrons flowing in the reverse direction from anode to cathode (x-ray tube require a DC current). Therefore, a certain electrical device called Rectifier must be provided for converting AC to DC by elimination of the negative half of the cycle.



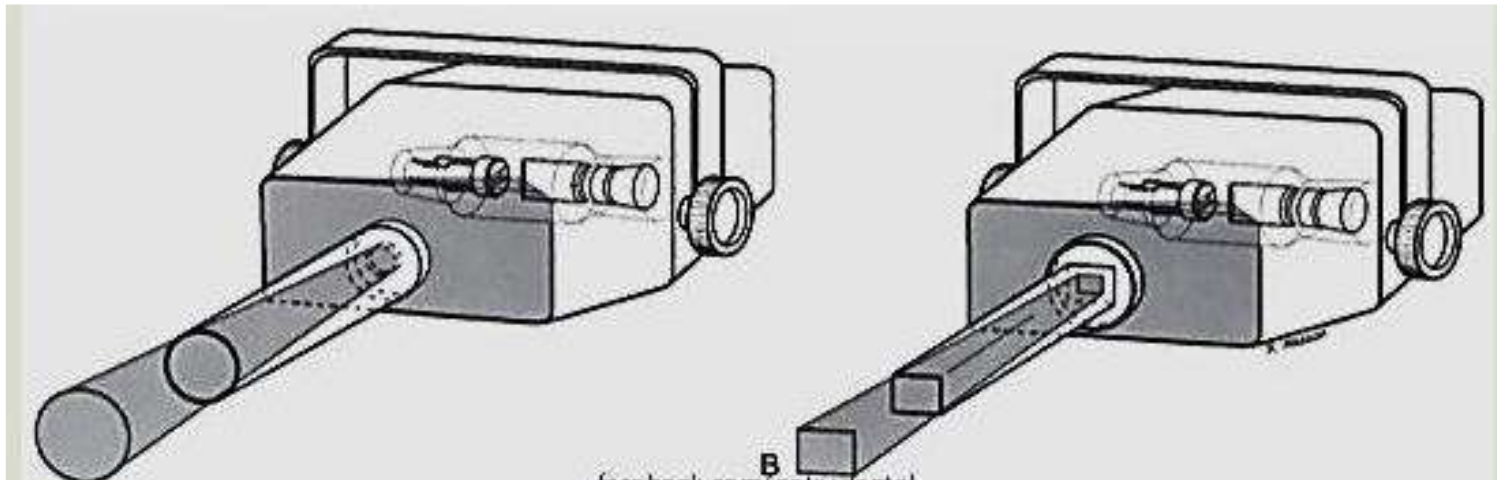
IV. Filtration

Low-energy (long wavelength) x-ray do not have enough energy to reach the film. To reduce patient dose, the less-penetrating photons should be removed. This can be accomplished by placing an aluminum filter in the path of the beam (added filtration).

Total filtration = inherent filtration plus external filtration (aluminum disks).

V. Collimation

- * **A collimator:** is a metallic barrier with an aperture in the middle used to reduce the size and shape of the x-ray beam and thereby the volume of irradiated tissue within the patient.



Inverse Square Law

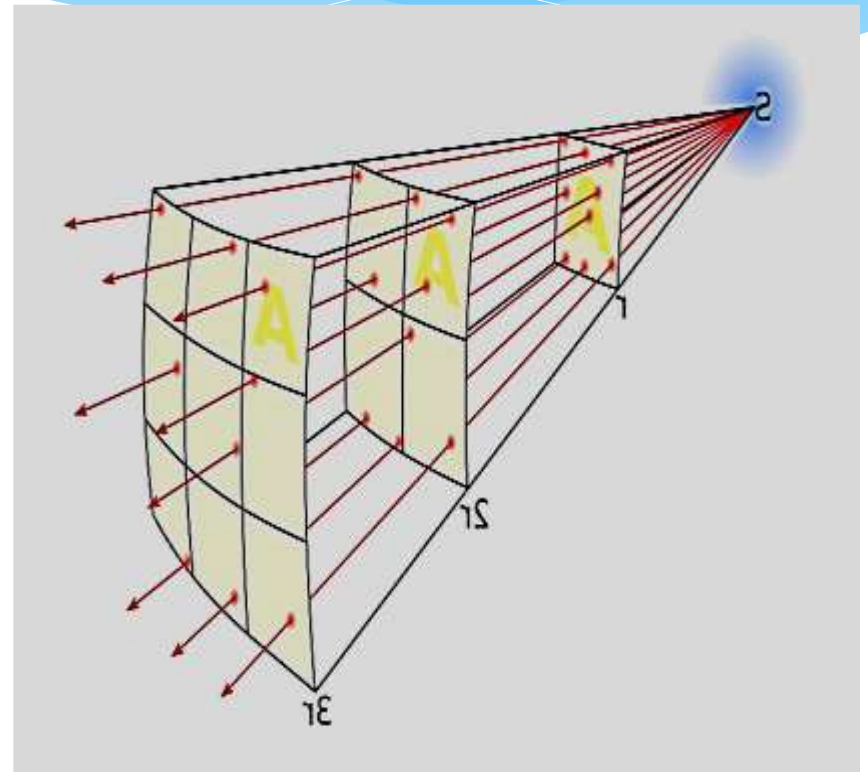
- * The intensity of an x-ray beam at a given point depends on the distance of the measuring device from the focal spot. For a given beam the intensity is inversely proportional to the square of the distance from the source.

Inverse Square Law

The Inverse Square Law

$$\frac{I_1}{(I_2)} = \frac{(d_2)^2}{(d_1)^2}$$

I_1 is the initial intensity of radiation, d_1 is the initial distance, and d_2 is the final distance, and I_2 is the final intensity.

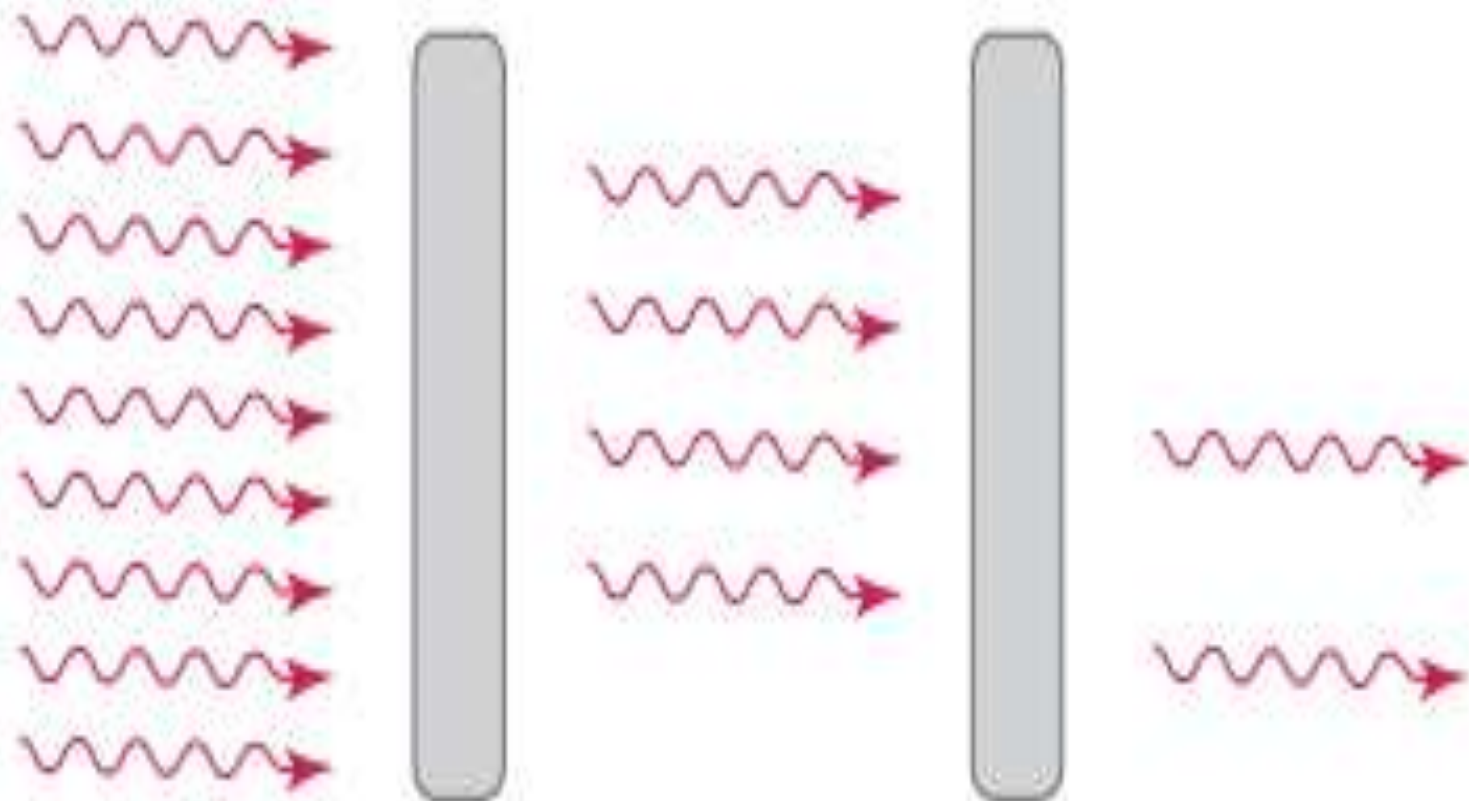


Definition of terms used in X – ray interaction:-

- * **Scattering:** - change in direction of photon with or without a loss of energy.
- * **Absorption:** - deposition of energy i.e. removal of energy from the beam.
- * **Attenuation:** - reduction in the intensity of X – ray beam caused by absorption and scattering
attenuation = absorption + scattering.
- * **Ionization:** - removal of an electron from neutral atom.

Half Value Layer

- * It's a method of monitoring the penetration quality of the X – ray beam. is done by placing thin filtering material such as **aluminum filter** in front of the beam so we continue increase the thickness of filtering material until we have a thickness that reduce the number of X – ray photons in the beam passing through it to **(one half)** In oral diagnosis the acceptable value is approximately 2 mm of aluminum



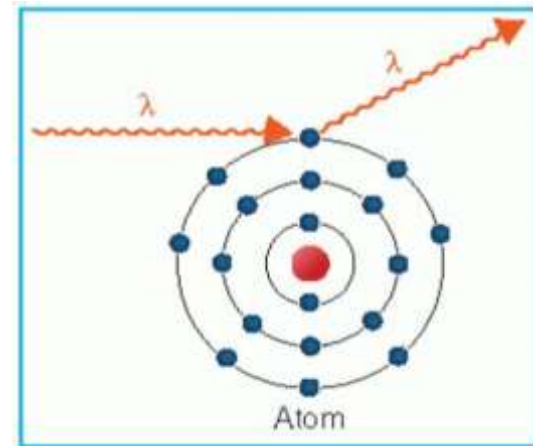
X-ray interaction with matter

(Absorption of X – ray)

x-ray are absorbed by any form of matter (**solid, liquid, and gas**), when photons reach an atom, different types of interaction may occur depend on photon energy:

- * 1. X-ray photons can pass through the atom without any change occurred to both of them.

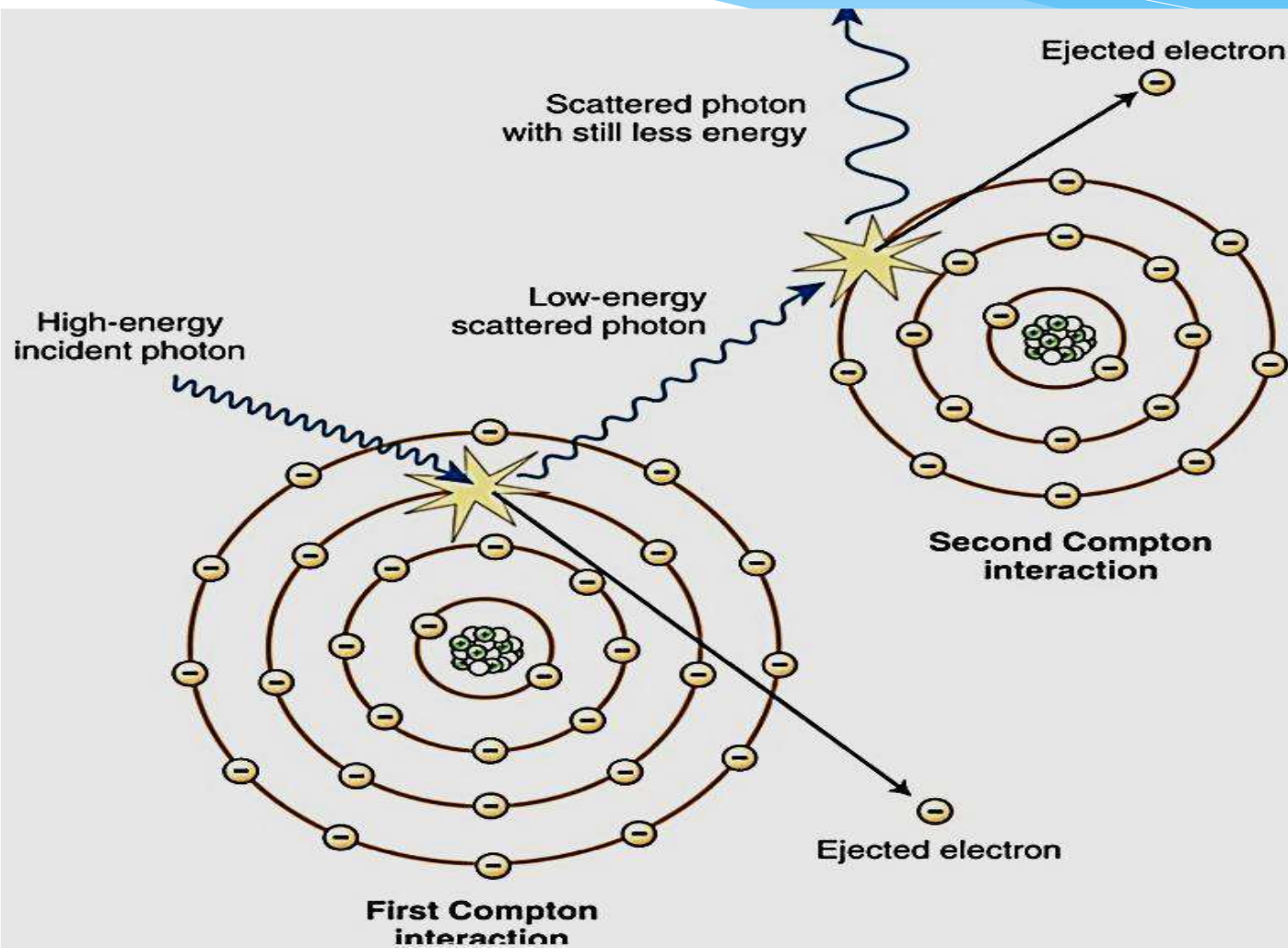
- * 2. **Coherent scattering** sometimes called **classical scattering** or **Thompson scattering** occur by interaction of low energy x-ray photon and atom .
- * There is no loss of photon energy only changes in direction (photon of scattered radiation)



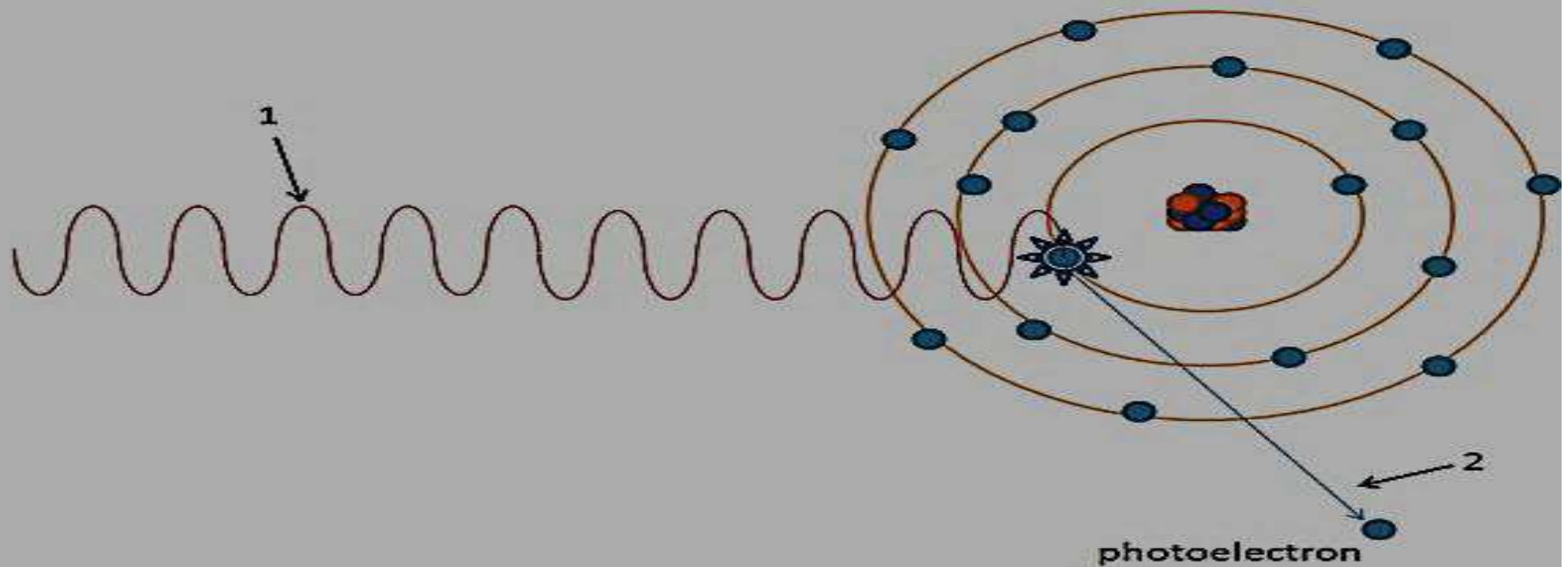


3. **Compton effect:** occurs when a photon interacts with an outer shell electron of atom. It results in ionization of atom, reduction of photon energy (there is some absorption of photon energy by ejected electron which undergoes further ionization interaction within the tissue) , and change in x-ray direction (scattered radiation).

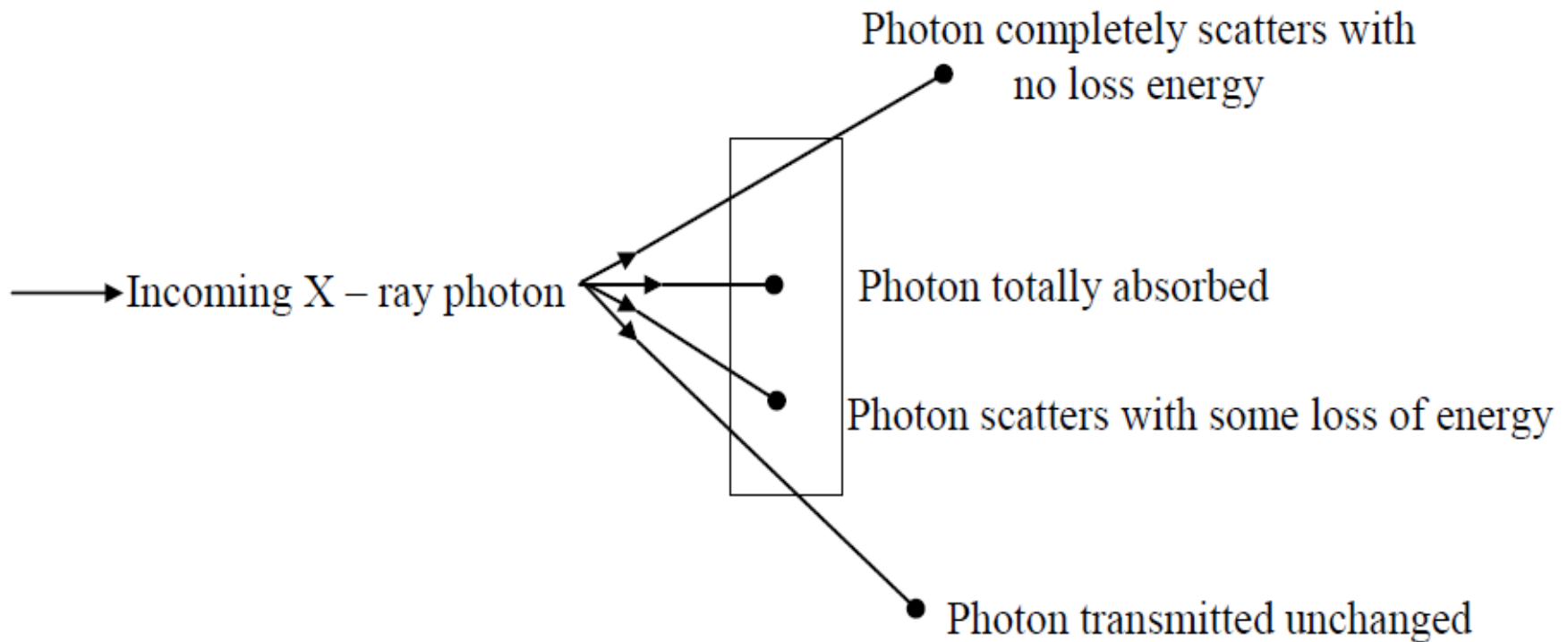
* **4. Photoelectric effect:** occur by X – Ray photon interaction with inner – shell electron of the tissue atom, the X – ray photon deposits all its energy this process is **pure absorption**. Now the inner – shell electron is ejected with considerable energy (now called a photo – electron) .So this high – energy ejected photo electron behaves like the original high energy X – ray photons interact and eject other electrons as it passes through the tissues.



Photoelectric effect



X-ray interaction with matter



X-ray measuring units:

- * **1.Traditional units**

- * **Roentgen**:- is the amount of x-radiation or gamma radiation which will produce in one cc of air ions.
- * **rad**: it is a roentgens absorbed dose .
- * **rem**: it is a roentgens equivalent man dose.
- * **RBE**: is a relative biological effectiveness dose.
- * **Curie**.

2.SI system

*Coulomb per Kilogram (**C/Kg**)

$$1\text{C/Kg} = 3876 \text{ R}$$

*Gray (**Gy**)

$$1\text{Gy} = 100 \text{ rad}$$

*Sievert(**Sv**)

$$1\text{Sv} = 100\text{rem}$$

*Becquerel(**Bq**)

$$1\text{Bq} = 2.7 \cdot 10^{-11} \text{ Ci}$$